Mohnad Waleed

Software Engineer

+20 1002573121 | mohndbata648@gmail.com | Mohand Waleed | LinkedIn| github.com/BataMohnd

EDUCATION

British University in Egypt — Bachelor of Science in Computer Science | Sep 2022 – Jun 2026

Relevant Coursework:
Data Structures • Algorithms Analysis • Software Methodology • Database Management • Artificial Intelligence • Internet Technology • Mobile Developing • Systems Programming • Computer Architecture • Data Mining & Warehouse

GRADUATION PROJECT

Fas7ny — AI-Driven Smart Travel & Leisure Planning Platform | *Flutter, Node.js, MongoDB, Redis, NLP, Machine Learning, Genetic Algorithms* | June 2026

- Built a full-stack AI mobile platform that autonomously assembles personalized, budget-optimized travel itineraries from a single conversational input (Arabic & English)
- Implemented 4 integrated AI modules: NLP engine (BERT-based NER + intent classification), Hybrid Recommendation Engine (content-based + collaborative + contextual filtering), Dynamic Pricing Optimizer (LP, IP, NSGA-II Genetic Algorithm), and AI Trip Planner (A* search, PDDL/POPF planning, CSP solver)
- Integrated Amadeus GDS API for live flight/hotel pricing and Paymob for PCI-DSS-compliant payment processing with JWT auth, MFA, and TLS 1.3
- Designed a microservices architecture (NLP, Recommendation, Planning, Pricing, Payment services) with MongoDB, Redis caching, and Firebase; targeting ≥10,000 concurrent users at 99.5% uptime
- Benchmarked algorithmic stack against 11 peer-reviewed papers (ViT+LSTM, NSGA-II, TravelAgent, DeepTravel, ATLAS); system achieves 92% preference accuracy and 95% budget adherence
- Supervised by Dr. Nermin Othman — Faculty of Informatics and Computer Science, BUE

PROJECTS

Car Rental Website | *React, Node.js, MongoDB (MERN Stack)* | April 2025

- Developed a full-stack car rental web application using the MERN stack with end-to-end functionality
- Implemented user authentication and authorization (Sign Up / Login) with secure session management
- Designed car listing and car details pages with booking functionality and date-based availability logic
- Built a reservation system with real-time pricing logic and integrated a payment workflow
- Developed an admin dashboard to manage cars, users, and reservations via RESTful APIs

Campus Club Management System – Chat Service | *React, Node.js, Express.js, REST APIs (Cloud Computing)* | 2026

- Designed and implemented the Chat Service module within a cloud-based campus club management platform
- Built protected admin message board with officer-only access using role-based authorization
- Developed REST endpoints (GET/POST /admin/chat/messages) with backend persistence and frontend integration
- Integrated the service into the broader microservice architecture with dedicated route namespacing
- Applied clean separation of concerns between frontend (AdminChat page) and backend API layer

Employee Attrition Analysis | *Python, Machine Learning, Power BI, Data Mining* | Dec 2025

- Analyzed a large-scale HR dataset (170K+ records) using a star schema (FactAttrition + dimension tables) to investigate employee attrition patterns
- Built an OLAP Cube in Power BI with DAX measures to explore attrition across education, department, tenure, and satisfaction dimensions
- Applied data preprocessing: encoding, scaling, and feature selection across two combined Kaggle datasets
- Implemented Association Rule Mining (Apriori, FP-Growth) and clustering (K-Means with PCA visualization)
- Built and evaluated classification models (Logistic Regression, KNN, Random Forest) prioritizing Recall to detect high-risk attrition

Automated Insulin Pump System | *Real-Time Systems, UML, Java* | April 2026

- Modeled a safety-critical real-time embedded system for automated blood sugar monitoring and insulin delivery
- Designed a time-driven architecture with a 10-minute primary monitoring cycle and 30-second self-test intervals
- Produced full UML suite: extended use case, class diagram, state machine, sequence, and collaboration diagrams
- Defined hard timing constraints and stimuli-response tables for all system events including safety overrides
- Applied real-time systems design rationale for a medical-grade application where correctness and timing are critical

E-Government Traffic System | *Java RMI, JDBC, Distributed Systems* | November 2025

- Developed a distributed e-government traffic unit system using Java RMI client-server architecture
- Implemented remote services for traffic violations inquiry, license renewal, and electronic payment
- Designed separate interfaces for citizens, employees, and administrators with role-based access control
- Applied secure payment logic, transaction validation, and penalty calculation with JDBC database integration
- Followed OOP principles and modular system design for maintainability and scalability

Smart Hospital Integrated Care & Resource Orchestration System (SH-ICROS) | *Formal Methods, VDM-SL* | March 2026

- Developed formal specifications (VDM-SL) for a hospital management system covering ED, ICU, OT, Pharmacy, and Diagnostic Services
- Defined 15 state invariants for clinical safety, referential integrity, and uniqueness across 7 entity types
- Modeled 6 staff roles with role-based permissions and 8 physical/human resource types with lifecycle states
- Specified 25+ operations including conflict detection, priority-aware resource allocation, and SLA escalation
- Designed UML class diagram as the direct basis for all VDM-SL type definitions and state declarations

Amusement Park Reservation System | *Java, OOP, Design Patterns* | June 2025

- Designed a full reservation system including tickets, parking, and services with eligibility checks and discounts
- Developed admin and staff dashboards to manage games and services using OOP and design patterns

4×4 Tic-Tac-Toe (AI) | *Python, Minimax, Alpha-Beta Pruning* |

- Built an advanced 4×4×4 Tic-Tac-Toe with AI opponent using Minimax with Alpha-Beta Pruning for optimized decision-making
- Designed state evaluation functions and focused on performance optimization and decision accuracy

Lara Game | *Unity, C#* | Oct 2024

- Developed a 2D story-driven adventure game in Unity with character movement, enemies, animations, and scene management

Library Management System | *Java* | January 2024

- Built a desktop library system with borrowing/returning, inventory management, and fee calculation using MVC-like structure

CERTIFICATIONS

- Flutter Development Course – Route Academy | January 2026

TECHNICAL SKILLS

Languages:	Python, Java, C, C#, JavaScript, SQL
Web:	HTML, CSS, React.js, Node.js, Express.js
Databases:	MongoDB, MySQL
Tools:	Git, GitHub, Linux, Jenkins, JUnit, WordPress, Power BI, Unity
Concepts:	OOP, REST APIs, MVC Architecture, Data Structures, Real-Time Systems, Formal Methods, Microservices